Chapter 7 Properties of Multiple Regression

Dennis Epple

Tepper School of Business

Copyright Dennis Epple

Statistical Decision Making (45-752)

[54]:
```
import tprstats
import pandas
import matplotlib.pyplot as plt
import numpy as np
from scipy.stats import t

# Jupyter "magic" command
# precision - round floats to 4 significant digits
%precision %.4f
```

[54]: '%.4f'

1 Properties of Multiple Regression

1.0.1 Learning objectives

After completing this chapter, you will be able to:

- Define the multiple regression model.
- Estimate a multiple regression model and interpret each of the coefficients.
- Understand the key condition to obtain unbiased coefficient estimates.
- Plot visual diagnostics that are useful in assessing the fit of the model to the data.
- Define a joint hypothesis and conduct a joint hypothesis test using the Wald test.
- Explain usefulness of plotting residuals against fitted values and independent variables.
- Undertake prediction and calculate a prediction interval.

1.1 Properties of Least Squares: The Gauss Markov Theorem

The multiple regression equation is presented below. The equation is linear in the population coefficients (betas) and has an additive error term ε_i.

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + ... + \beta_K x_{Ki} + \varepsilon_i$$

In our discussion of the simple regression model, we introduced notation for the estimated coefficients, predicted values, and estimated error terms. The following extends this notation to the multiple regression model. The estimated coefficients are denoted b_0, b_1,...,b_K.

Using these estimated coefficients, we obtain the predicted value for each observation:

$$\hat{y}_i = b_0 + b_1 x_{1i} + b_2 x_{1i} + ... + b_K x_{Ki}$$

We then calculate the estimated error term, e_i, for each observation:

$$e_i = y_i - \hat{y}_i$$

Brief recap:

- β_k denotes a parameter (i.e., an unknown population coefficient) and b_k denotes an estimated coefficient.
- ε_i denotes the error term for observation i in the population model.
- e_i denotes the estimated error term for observation i.

As discussed in Chapter 4, a fundamental result in statistics is the Gauss-Markov Theorem. The assumptions employed in proving the Theorem for the multiple regression model are stated below. This is followed by interpretation that builds on the discussion in Chapter 4.

Suppose the following assumptions are satisfied:

1. The ε_i are distributed independently of the x_{ik} for all observations i for all variables $k = 1, ..., K$.
2. The ε_i are from a population with a mean of zero.
3. The ε_i are from a population with a constant variance σ^2: $E(\varepsilon_i) = 0$, $Var(\varepsilon_i) = \sigma^2$.
4. The error terms are not correlated with each other: $Cov(\varepsilon_i, \varepsilon_j) = 0$ for all $i \neq j$.

The **Gauss Markov Theorem** establishes that, under the above assumptions, least squares is the Best Linear Unbiased Estimator of the coefficients. This often goes by the shorthand that least squares is BLUE.

Unbiased means that the mathematical expectation of the estimated coefficient equals the population coefficient: $E(b_k)$=β_k for all coefficients $j = 0, 1, ..., K$.

Intuitively, what does "Best" mean in this context? Think about it as follows. You have a set of data, values of x_{ik} and y_i. You are going to use a linear unbiased estimator. There are many linear unbiased estimators.[1] Least squares is used extensively in practice. Why? Under the conditions assumed in the Gauss-Markov Theorem, there is no other linear unbiased estimator that outperforms least squares in precision of estimating the coefficients.

Here is an example to illustrate randomness of the x_i in a regression. Suppose I wish to investigate how the natural gas used to heat my home in the winter varies with outside temperature. I decide to obtain data on average outside temperature for each day for the next 30 days. Call these value x_i for $i = 1, ..., 30$. Suppose that I record the amount of natural gas used in my home for the next 30 days. Call these value y_i for $i = 1, ..., 30$. Having collected these data, I can estimate a regression to

[1] For example, suppose we randomly select half of the observations in our data and throw those observations away. We then estimate a regression with the remaining observations. It would be a linear unbiased estimator, but obviously not the best we can do with the data.

investigate how natural gas usage in my home varies with outside temperature. Of course, outside temperature will not perfectly predict natural gas usage in my home. Natural gas usage will also depend on whether the sun is shining during the day, whether the wind is blowing, opening and closing of doors to the outside, natural gas used for cooking, and so forth. In this application, the x_i, the outside temperatures, are random. The other influences on natural gas usage just listed above give rise to the error terms ε_i in the model.

As discussed in Chapter 3, Assumption 1 of the Gauss-Markov theorem is the most important assumption for applying regression. Here are two examples of why the first assumption is so important.

Example 1: Consider forest fires in California. Housing construction entails a variety of human activities that may create sparks that inadvertently ignite forest fires. Such activities include extension of roadways, transportation of materials, operation of both gasoline and electrically powered equipment, and, in some instances, cigarette smoking. Suppose that, given the tremendous damages from forest fires, the state government of California is considering imposition of restrictions on housing construction.

Suppose your consulting company is asked to do an assessment of the extent to which housing starts affects forest area burned, and you are assigned the job. You find that annual data are available for the past 34 years. Suppose you gather annual data over the past 34 years on forest area burned, F and housing starts H in California. You then estimate a regression with F as the dependent variable and H as the independent variable. Would this provide a valid estimate of the effect of housing starts on forest area burned? Probably not. Suppose other variables affect forest area burned such as ambient temperature and precipitation. The effects of those variables are then impounded in the error term in the regression equation. Suppose those other variables are correlated with housing starts because housing starts are more likely to occur during times of warm weather and low precipitation. Then, when we regress area burned on housing starts, the coefficient of housing starts will be biased because housing starts will also impound part of the effects of temperature and precipitation. Thus, it is important to include all relevant variables when estimating a regression model. Of course, you cannot look up a menu to find which variables should be in the regression. You must rely on your knowledge of the problem to determine which variables should be included.

Example 2: Suppose your consulting firm is hired by a company to investigate the effect of advertising expenditure on sales. The company has data for the past 60 months on the dollar value of advertising A and the dollar value of sales S. Consider the following regression of sales on advertising:

$$S_i = \beta_0 + \beta_1 A_i + \varepsilon_i$$

Would this regression provide a meaningful estimate of the effect of a change in advertising on sales? Suppose you talk with the CEO and find that the company has generally tended to increase advertising in months where sales revenue was running short of expectation and cut back in months when sales revenue was running above expectations. This tells us that the company increases A_i in months when $\varepsilon_i < 0$ and decreases advertising in months when $\varepsilon_i > 0$. This creates a negative correlation between A_i and ε_i. Hence, the assumption that the ε_i are independent of the A_i is violated. Intuitively, the estimated coefficient b_1 will be a downwardly biased estimate of the true coefficient β_1. This is a situation with two-way causality. If advertising is effective, then

advertising increases sales. However, if advertising is increased in inverse relationship to sales, then the estimated slope coefficient will understate the true effect of advertising. We must then adapt the approach accordingly. Thus, the above regression will tend to understate the actual effectiveness of advertising. Simply estimating the above regression will not do. In this instance, it would be natural to ask the CEO to provide data on the amount by which advertising was adjusted in response to sales.

Example 3: The first assumption may also be violated if the inappropriate functional form is chosen for one or more variables. For example, suppose the appropriate functional form for a variable is the logarithm of a variable, but the variable is not expressed in log form. We will see in a later chapter strategies for choosing functional form.

There are many applications in which one or more right-hand-side variables may not be random. For example, we study a model of pricing of diamonds below. In this case, the characteristics of the diamond, clarity and carat, are not randomly determined. In such cases, e still need to be careful to include all relevant right-hand-side variables. We will investigate the reason in the analysis below of the diamond data.

The above detailed discussion of the first assumption is because it is the most important assumption. We now discuss the remaining assumptions.

The second assumption will be trivially satisfied if you include an intercept. It is rare that you would have a model in which you do not include an intercept.

Suppose the third assumption is not satisfied. This is called heteroskedasticity. If we have heteroskedasticity and fail to address the issue, the standard errors of our estimated coefficients will be biased. This may result in incorrect finding regarding statistical significance of the coefficients. Fortunately, this is easily addressed by routinely correcting for the possible presence of heteroskedasticity as discussed in Chapter 4.

It is frequently the case that the fourth assumption does not hold when we estimate a regression using time series data. If the assumption is violated and we do not address the problem, we will have biased standard errors and may make incorrect decisions about statistical significance of the coefficients. There are also situations where violation of assumption four may cause biased estimates of coefficients. We will see how to address this issue in detail in a later chapter on estimating regression models with time series data.

1.2 Analyze Data for 211 Diamonds

As an application, we will next study how Prices of diamonds vary with Carat weight and Clarity. This application will help improve your familiarity with use of multiple regression. It will also serve to illustrate the effect of leaving out a variable that should be in the model. In addition, this application paves the way to study of functional form.

We will use data for 211 diamonds. Import the Excel data file Diamonds_211.xlsx.

[55]:
```
Diamonds_211 = pandas.read_excel("data/Diamonds_211.xlsx")
```

Let's begin by "looking at the data". We'll first first view summary statistics. We see that diamonds range in price from $2,056 to $28,546. We see that half of these diamonds sold for $6,914 or more.

- What is the range of weight (Carat) for these diamonds?

- Fill in the following blank: Half of these diamonds weighed _____ carats or less.
- When you have chosen your answers to the previous two questions, check against the solution in the footnote.[2]

We also see that these diamonds all have the same Cut and Color. We will use the Salesperson variable in a later chapter. You can ignore that variable for the time being.

[56]:
```
Diamonds_211.describe()
```

[56]:

	Price	Carat	Clarity	Color	Cut	Salesperson
count	211.000000	211.000000	211.000000	211.0	211.0	211.000000
mean	8769.364929	1.239573	4.270142	11.0	2.0	5.208531
std	5744.400127	0.511450	1.206336	0.0	0.0	2.469649
min	2056.000000	0.510000	2.000000	11.0	2.0	1.000000
25%	4037.500000	0.800000	3.000000	11.0	2.0	3.000000
50%	6914.000000	1.110000	4.000000	11.0	2.0	5.000000
75%	11750.500000	1.585000	5.000000	11.0	2.0	7.000000
max	28546.000000	2.470000	8.000000	11.0	2.0	9.000000

1.2.1 Calculate and Think About The Correlation of Carat and Clarity

Calculate the correlation.

[57]:
```
# Note: This is called the Pearson Correlation Coefficient
Diamonds_211["Carat"].corr(Diamonds_211["Clarity"])
```

[57]: -0.4331

Is it surprising that the correlation is negative? No, it is not surprising. It is much easier for a jeweler to cut a small diamond to avoid including imperfections than to cut a large diamond while avoiding inclusion of such imperfections. Hence, larger diamonds tend to have more imperfections than smaller diamonds, giving rise to the negative correlation of Carat and Clarity.

1.2.2 Viewing and Interpreting Histograms

As you know, viewing histograms of variables is another useful way to look at your data. Below are two histograms of the Price variable. The histogram on the top left is plotted using the default settings in the Pandas package. Pandas plots are created using the matlotlib package and thus share the same functionality as matplotlib plots.

This example places multiple graphs together in a single figure using matplotlib. A figure in matplotlib is divided into axes with each axis generally containing one subplot. The plt.subplots(nrows=2, ncols=2) specifies a 2x2 figure. To indicate what axis a plot appears in, we write ax=axes[0, 0] which corresponds to the axis in 1st row and 1st column of the figure. Why 0 instead of 1? Python counts starting from 0 rather than 1.

Sometimes you may want a more fine-grained histogram. The **bins** feature illustrated in the second histogram command accomplishes this. Depending on your data, you may wish to try various values of breaks until you get a histogram that you think provides a good portrayal of your data.

[2] Carat weights range between .51 and 2.47. Half of the diamonds have carat weight of 1.11 or less.

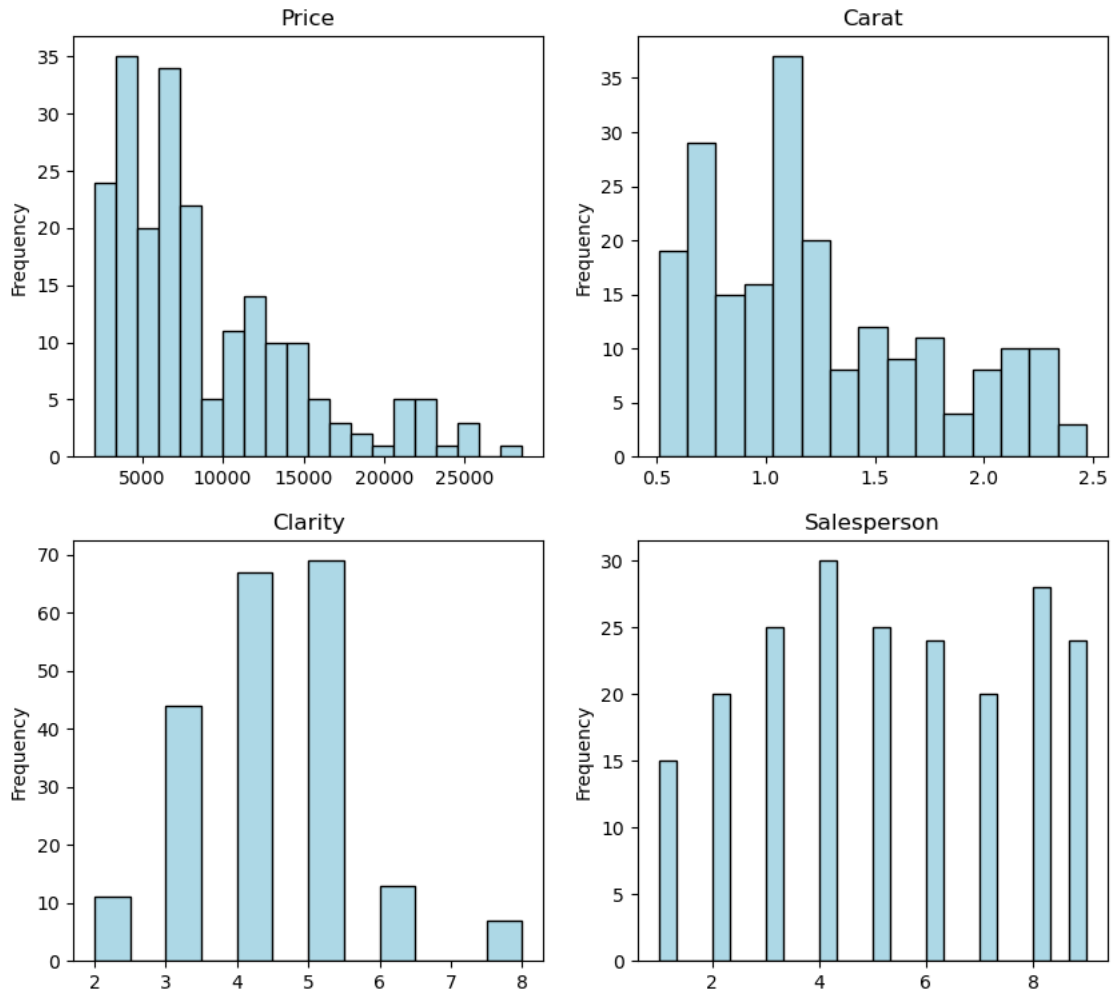
We see that the distribution of Price is **skewed to the right**. There are a few diamonds with very high prices and more diamonds with lower prices. Please remember this terminology. If one tail of a distribution is longer and thinner than the other, we say it is **skewed**. If the distribution of a variable has a long narrow left tail, we say it is **skewed to the left**.

From the histogram for Carat, we see there is some right skewness, though less pronounced than for price. The histogram command for Carat shows you how to add color. Try changing the command to show color "red". Then try "lightblue". Then guess the name of another color and give it a try. If you get an error message, it means that the color you chose is not part of the matplotlib color palette.

From the histogram for Clarity, we see that diamonds with clarity 4 and 5 are more numerous than diamonds with other clarity values. The histogram command for Clarity shows you how to add a title for the horizontal axis, a main title, color, and bins = 10. Note from the histogram that there are no diamonds with Clarity=7 in our dataset. This further illustrates the usefulness of viewing a histogram of the variables.

From the histogram for Salesperson, we see the number of diamons sold by each of the nine salespeople.

[58]:
```
fig, axes = plt.subplots(nrows=2, ncols=2)
Diamonds_211["Price"].plot.hist(ax=axes[0, 0], title="Price", figsize=(10,␣
↪9),bins=20,edgecolor="black",color="lightblue")
Diamonds_211["Carat"].plot.hist(ax=axes[0, 1], bins=15, title="Carat",␣
↪figsize=(10,9),edgecolor="black", color="lightblue")
Diamonds_211["Clarity"].plot.hist(ax=axes[1, 0], bins=12, title="Clarity",␣
↪figsize=(10,9),edgecolor="black",color="lightblue")
Diamonds_211["Salesperson"].plot.hist(ax=axes[1,1],bins=24,␣
↪title="Salesperson",figsize=(10,9),edgecolor="black",color="lightblue")
plt.show()
```



1.3 Estimate a Regression Using Diamond Data

Now let's estimate a regression of Price against Carat and Clarity. Before looking at the command below, see if you can write the command to do the regression. Name your regression RegDiam211.

When you have estimated a regression, you will want to ask yourself the following six questions. We have previously discussed the first three of these questions.

1. Do the coefficients of the right-hand-side variables have the expected algebraic sign?
2. Are the coefficients statistically significant at the 5% level?
3. Do the magnitudes of the significant coefficients seem plausible?
4. Have you chosen an appropriate functional form?
5. Have you included all relevant variables in your regression?
6. Is there a potential issue of two-way causality?

Let's discuss each of above in the context of our diamond model:

1. We expect positive coefficients for Carat and Clarity. So the answer to question 1 is yes.

2. We see from the regression output that the p-values for Carat and Clarity are both less than .05. Hence, the coefficients are statistically significant at the 5% level. There are situations where you may choose to include variables even if they are not significant at the five percent level. For example, this was the case for the Tread variable in the tire pricing model.
3. To answer this requires expertise about the diamond market. In future courses in Tepper and in your future careers, you will typically be working on a question about which you have gained considerable expertise. Hence, you will typically be able to answer question 3 for your applications of regression.
4. We will study how to evaluate functional form in a future chapter. We will find that the regression above is not a very good functional form for the data and we will find a better functional form. However, you would often begin with a linear model, as we are doing with this application.
5. When you develop a regression model, you will need to think very carefully about the variables that should be included in your regression model. In this particular application, the answer is easy. These diamonds vary in only two characteristics, Carat and Clarity. In most applications you will undertake, the choice is not so straightforward. We will see more challenging applications as we proceed in the course.
6. I do not see a reason to think that two-way causality is an issue in this application.

[59]:
```
RegDiam211 = tprstats.model("cs", "Price~Carat+Clarity", Diamonds_211)
RegDiam211.summary()
```

[59]:

Dep. Variable:	Price	**R-squared:**	0.950
Model:	OLS	**Adj. R-squared:**	0.950
No. Observations:	211	**F-statistic:**	773.8
Covariance Type:	HC1	**Prob (F-statistic):**	4.54e-97

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
Intercept	-1.16e+04	759.549	-15.278	0.000	-1.31e+04	-1.01e+04
Carat	1.193e+04	314.094	37.973	0.000	1.13e+04	1.25e+04
Clarity	1308.8177	124.664	10.499	0.000	1063.051	1554.584

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)

The regression output above converts the coefficients to scientific notation. To see more digits we can use the params option as shown below to see more detail. We see that a one Carat increase in size increases predicted price by $11927 holding Clarity constant. A one unit increase in Clarity increases predicted price by $1308 dollars holding Carat constant. You may be puzzled by the negative intercept. Don't worry about this now. We will return to this issue in a later chapter

[60]:
```
RegDiam211.params
```

[60]:
```
Intercept   -11604.045456
Carat        11927.145390
Clarity       1308.817663
dtype: float64
```

For your practice: Consider two diamonds. Both have Carat=.75, but one has Clarity=4 and the other has Clarity=5. The regression predicts that the diamond with Clarity=5 will sell for $1,309

more than the diamond with Clarity=4. Do you agree?

1.3.1 Calculate and Interpret Standardized Coefficients and Elasticities

We see from the standardized coefficients below that a one standard deviation increase in Carat increases predicted price by 1.06 standard deviations. A one standard deviation increase in Clarity increases predicted price by 0.27 standard deviations. The coefficient of a variable depends on units of measurement. Suppose we measured weight of the diamonds in milligrams instead of carats. There are 200 hundred milligrams per carat. The coefficient of milligram would be the coefficient of carat divided by 200: 11927.1/200=59.63. The standardized coefficient of milligram would be the same as the standardized coefficient of Carat. The advantage of standardized coefficients is that they do not depend on scaling of the variables.

[61]:
```
RegDiam211.scaled_coefficients()
```

[61]:

	coefs	std_coefs	elasticities
Carat	11927.145390	1.061928	1.6859
Clarity	1308.817663	0.274854	0.6373

From the elasticities, we see that a 1% increase in Carat increases predicted price by 1.69 percent. Elasticity is also a scale free measure. If we measured weight in milligrams rather than carats, the elasticity would be the same. A 1% increase in Clarity increases predicted price by 0.64%. Recall that clarity is an integer variable rather than a continuous variable. Thus, it is not possible to increase Clarity by 1%, but the elasticity still gives us a sense of responsiveness of Price to Clarity relative to responsiveness of Price to Carat.

1.3.2 Calculating Confidence Intervals for Coefficients

The confidence intervals appear in the summary output. You can also access the confidence functions using the `conf_int` function.

[62]:
```
RegDiam211.summary()
```

[62]:

Dep. Variable:	Price	**R-squared:**	0.950
Model:	OLS	**Adj. R-squared:**	0.950
No. Observations:	211	**F-statistic:**	773.8
Covariance Type:	HC1	**Prob (F-statistic):**	4.54e-97

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
Intercept	-1.16e+04	759.549	-15.278	0.000	-1.31e+04	-1.01e+04
Carat	1.193e+04	314.094	37.973	0.000	1.13e+04	1.25e+04
Clarity	1308.8177	124.664	10.499	0.000	1063.051	1554.584

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)

Let's view the 95% confidence intervals for the regression coefficients using the `conf_int` command. This command shows us more digits that are shown in the above regression output. We see that the confidence interval for the coefficient of carat is [11308, 12546]. This tells us there is a 95% probability that the interval [11308, 12546] contains the unknown population coefficient for Carat.

[63]:
```
RegDiam211.conf_int()
```

[63]:
```
                       0             1
Intercept -13101.446712 -10106.644201
Carat      11307.929061  12546.361720
Clarity     1063.051287   1554.584039
```

In your future work, you may read a report in which a coefficient and its standard error are reported, but the confidence interval is not reported. Hence, it is important for you to be able to calculate a confidence interval yourself. Let's calculate a 95% confidence interval for the coefficiet of Carat in our diamond regression. back to our regression, we see that the estimated coefficient is 11927.0 and the estimated standard error is 314.1.Looking For our calculating, we need the t-statistic for 208 degrees of freedom. The degrees of freedom is the difference between the number of observations and the number of coefficients we estimated: 211-3=208. The t-statistic is calculated using the command in the cell below. We see that the t-statistic is 1.97. Hence, our confidence interval is:

$$[11927 - 1.97 * 314.1, 11927 + 1.97 * 314.1] = [11308, 12546]$$

Interpretation: There is a 95% probability that the interval above contains the population coefficient of carat. Look back to the results of our command `RegDiam211.conf_int()` and you will see that, within round-off error, the confidence interval for Carat is the same value as we just calculated by hand. For practice, calculate the confidence interval for Clarity. Then check your answer by looking back at the results of the `RegDiam211.conf_int()` command.

[64]:
```
t.ppf(0.975, 208)
```

[64]:
```
1.9714
```

1.4 A Convenient Command for Testing A Hypotheses about One or More Coefficients

Suppose a jeweler believes that the population coefficient for Clarity is $12,500. We can look at the confidence interval just above and see that 12,500 lies inside the confidence interval. Hence, we do not reject this null hypothesis at significance level .05.

Suppose the jeweler would like us to test the hypothesis at the 10% level. We can, of course, do the calculation of the 10% confidence interval by hand, but that is a bit tedious. The following command provides a very convenient way to obtain the p-value for the jeweler's hypothesis. We obtain p-value=.07. This p-value is less that .10. Hence, we reject the null hypothesis at a 10% significance level.

[65]:
```
RegDiam211.wald_test("Carat = 12500")
```

```
p-value:  0.0696
```

Suppose the jeweler believes the population coefficient of carat is $12,500 and the population coefficient of Clarity is $1,200. Let's test this hypothesis using $\alpha = .05$. We obtain p-value=.023. Hence, we reject this null hypothesis. This is a very useful command in many applications. It is called the Wald test after its inventor, Abraham Wald. It will become one of your favorite

commands. Be sure to study the format of the command. When we test a hypothesis about two or more coefficients, it is called a **Joint Hypothesis Test**.

[66]:
```
RegDiam211.wald_test("Carat=12500, Clarity=1200")
```

```
p-value:  0.0227
```

1.5 Using Multiple Regression for Prediction

A jeweler has obtained a new diamond with Carat equal to 0.75 and Clarity equal to 8. The jeweler would like to have a prediction of the price that the diamond will sell for and a 95% prediction interval. We calculate the prediction interval using the command below. The model predicts that the diamond will sell for $7,812 with a 95 prediction interval [$5140, $10483].

[67]:
```
# Predict price of a diamond with Carat=.75 and Clarity=8 and calculate a 95%␣
↪prediction interval.
Xnew = pandas.DataFrame({"Carat": [0.75], "Clarity": [8]})
RegDiam211.prediction_intervals(Xnew, alpha=0.05)
```

[67]:

	Predicted	Lower	Upper
0	7811.854891	5140.122814	10483.586968

Recall why we call this a prediction interval instead of a confidence in interval. There are two sources of uncertainty when we do a prediction. We don't know what error term will arise for a Diamond when it is sold. This error term will depend on the relative bargaining ability of the customer relative to the salesperson. We also do not know the population coefficients in the regression model, and we have to estimate them. A prediction interval takes account of both of these sources of uncertainty.

The jeweler has obtained three new diamonds and would like to obtain the predicted price and the 95% prediction interval for each of them. The first has Carat=.8 and Clarity=8. The second has Carat=1.3 and Clarity=6. The third has Carat=2.5 and Clarity=2. We can do the calculations for all three of the diamonds as follows. We create a pandas dataframe to store the new values, Xnew, and use the `prediction_intervals` command to calculate the prediction intervals. The predictors must be provided as a pandas dataframe.

Be sure you know how to interpret these results. Look back to the interpretation of the prediction we did above if you need to review to interpretation of the results.

[68]:
```
Xnew = pandas.DataFrame({"Carat": [0.8, 1.3, 2.5], "Clarity": [8, 6, 2]})
RegDiam211.prediction_intervals(Xnew, alpha=0.05)
```

[68]:

	Predicted	Lower	Upper
0	8408.212161	5733.071849	11083.352472
1	11754.149530	9164.138653	14344.160407
2	20831.453346	18177.742958	23485.163733

Suppose a jeweler has obtain 50 new diamonds and would like to obtain predicted prices and a 95% prediction interval for each. The jeweler automates the proces by putting the information in an Excel spreadsheet with "Carat" and "Clarity" as headers. The jeweler does so and names the spreadsheet NewDiamonds.xlsx. The jeweler then imports the data file and executes the following

commands to obtain predicted prices and 95% prediction intervals for each new diamond. As you will see, the jeweler chose the name PredNewDiamonds for the output.

[69]:
```
NewDiamonds = pandas.read_excel("data/NewDiamonds.xlsx")
```

[70]:
```
NewDiamonds.head()
```

[70]:

	Carat	Clarity
0	1.70	2
1	1.08	3
2	1.13	4
3	0.58	5
4	0.55	6

[71]:
```
# Calculate predicted prices and 95% prediction intervals for 50 new diamonds
PredNewDiamonds = RegDiam211.prediction_intervals(NewDiamonds, alpha=0.05)
# Round predicted prices to whole dollar amounts and view results for the first␣
↪few diamonds
PredNewDiamonds.head().round()
```

[71]:

	Predicted	Lower	Upper
0	11290.0	8705.0	13875.0
1	5204.0	2648.0	7759.0
2	7109.0	4571.0	9647.0
3	1858.0	-692.0	4408.0
4	2809.0	243.0	5374.0

To view the entire set of results, we can write the results to a spreadsheet. To do so, let's combine the predicted values with the Carat and Clarity data for the new diamonds. Let's name the result NewDiamonds_and_Predictions. The first command in the following code merges that Carat and Clarity data with the predictions. The second command rounds off the data to two digits after the decimal point. The third command puts the results into a file named output.csv. The fourth command downloads the data.

[72]:
```
# Combine the data for new diamonds with the predicted prices
NewDiamonds_and_Predictions = pandas.concat([NewDiamonds, PredNewDiamonds],␣
↪axis=1)
# Round the results to two digits
NewDiamonds_and_Predictions=NewDiamonds_and_Predictions.round(2)
# Put the results in a csv file named Results.csv
NewDiamonds_and_Predictions.to_csv("Results.csv",index=False)
```

1.5.1 Suppose We Leave Out A Variable That Belongs In Our Model

Let's investigate the result we obtain if we estimate a model with Clarity but without Carat. This will provide you intuition about why it is important to include all relevant variables. In the regression output below, we see that the coefficient of Clarity is negative and statistically significant. What is the intuition?

We saw above that there is a negative correlation between Carat and Clarity. Large diamonds have lower clarity than small diamonds. Nonetheless, large diamonds tend to be more valuable than small diamonds. When we leave out the Carat variable, the Clarity variable attempts to do double duty. The coefficient of Clarity captures the direct effect of Clarity on Price and part of the effect of Carat on Price. The latter effect dominates the former in this application, resulting in a negative coefficient on Clarity. The coefficient of Clarity is biased downward. Of course, it is implausible that a variable as important as Carat would be left out of a regression, but this example clearly illustrates the bias that can arise if a relevant variable is left out. Be sure to think through this example until you get the intuition for why the coefficient of Clarity changes sign when we leave out Carat.

[73]:
```
# Regress Price against Clarity. Name the result RegNoCarat
RegNoCarat = tprstats.model("cs", "Price~Clarity", Diamonds_211)
RegNoCarat.summary()
```

[73]:

Dep. Variable:	Price	**R-squared:**	0.034
Model:	OLS	**Adj. R-squared:**	0.030
No. Observations:	211	**F-statistic:**	9.984
Covariance Type:	HC1	**Prob (F-statistic):**	0.00181

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
Intercept	1.253e+04	1123.542	11.155	0.000	1.03e+04	1.47e+04
Clarity	-881.2864	278.917	-3.160	0.002	-1431.138	-331.435

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)

We now return to study of our model, RegDiam211, that includes both Carat and Clarity.

1.6 View a Plot of The Data and The Regression Plane

When you have estimated a model, you will want to investigate how well your model fits the data. For example, might there be a nonlinear pattern in the data that is not captured by the model? Below is a three-dimensional plot of Price against Carat and Clarity. By trial and error, I found angles for viewing along the regression plane. From the plot, we observed a pattern with negative residuals near the middle and positive residuals near the ends. We under-predict for large and small diamonds and over-predict for diamonds of intermediate size. This suggests that the linear model does not fit the data well. We will return to this issue in our next chapter, which covers choice of functional form.

1.7 Diagnostics for a Regression Model

Suppose we have three explanatory variables. We cannot do a four-dimensional plot to view the regression and data points. This applies more generally with more than two explanatory variables. Hence, we need diagnostics that can be applied in situations with more than two right-hand side variables. Such diagnostics are presented below.

1.7.1 Calculate Fitted Prices and Residuals

In the diagnostics below, we will utilize the fitted values and residuals from the regression. Here is a brief reminder of the meaning of fitted values. Let i denote an observation from our diamond data data set. Let $Carat_i$ and $Clarity_i$ denote the explanatory variables for observation i. The fitted value for observation i is then:

$$\hat{y}_i = -11604 + 11927.0 * Carat_i + 1308.8 * Clarity_i$$

Consider the first observation $i = 1$. Look at the Diamond_211 data set and you will see that $Carat_1 = 1.7$ and $Clarity_1 = 2$. Plugging these values into the above equation, we obtain the fitted price for this diamond is $\hat{y}_1 = 11289$.

When we issue the predict command, it does the calculation for every diamond in the data set. Let's issue the fitted command and name the fitted values yhat. After we issue this command, let's look at the first value of yhat. We do this with the yhat[0] command. We see that the value rounds to 11289, as we calculated just above.

[74]:
```
# Calculate fitted values and name the result yhat
yhat = RegDiam211.predict()
yhat[0]
```

[74]:
```
11289.7370
```

The estimated error term for observation i is the difference between the price for which the diamond sold and the fitted price:

$$\hat{e}_i = Price_i - \hat{y}_i$$

The first diamond in the data set sold for \$11,102. Hence, the estimated error term for the first observation is: \$11102 - \11289 =
-187. This tells us that model over-predicts the price of the first diamond by $187.

Below I used the resid command to obtain the estimated error terms for all diamonds in the data. I named the result resid. The resid[0] command reports the first error term. After you run this command, look at resid and you will find that the first value of resid is -187, as we calculated just above.

[75]:
```
resid = RegDiam211.resid
resid[0]
```

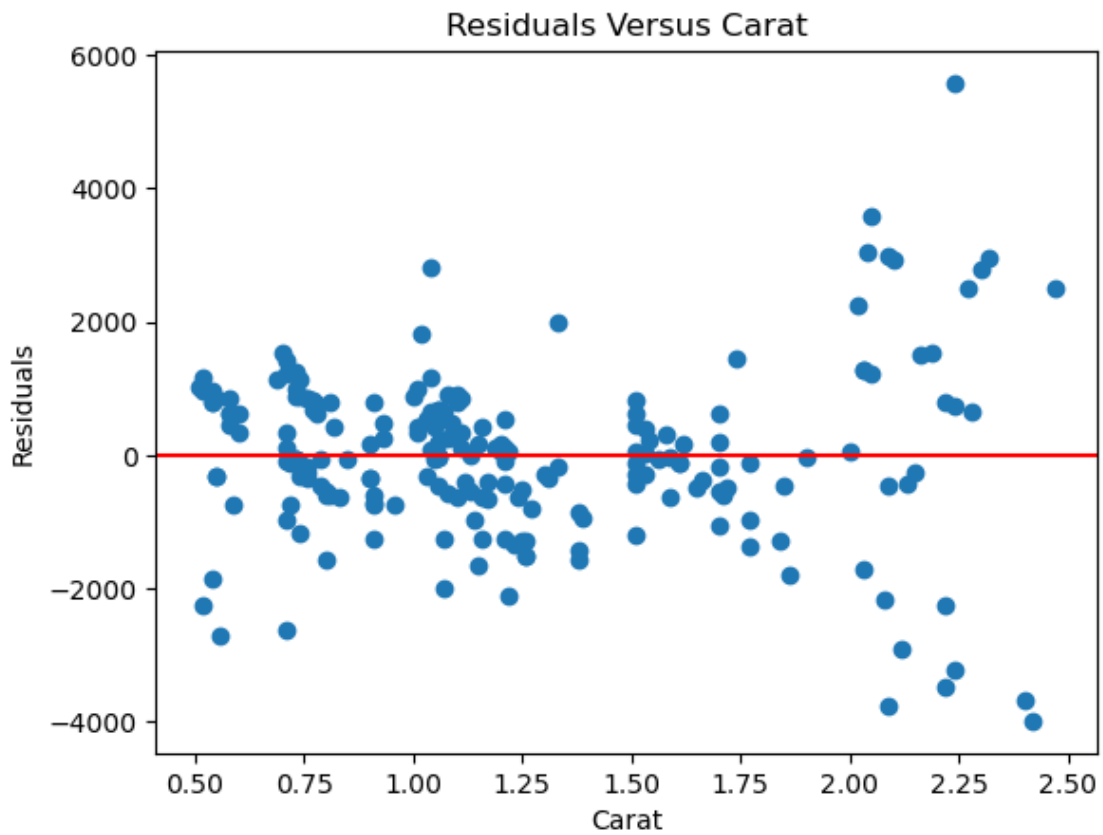
[75]:
```
-187.7370
```

1.7.2 Plot Residuals Against Explanatory Variables

It can often be instructive to plot the residuals against each explanatory variable. From the following plot of residuals against Carat, we see clear evidence of heteroskedasticity. There is much more variance in the residuals for large diamonds than for small diamonds. We already addressed the possible presence of heteroskedasticity in `tprstats.model`. Hence, we do not need to do anything more with respect to heteroskedasticity.

[76]:
```
plt.scatter(Diamonds_211["Carat"], resid)
plt.axhline(0, color="red")
plt.title("Residuals Versus Carat")
plt.xlabel("Carat")
plt.ylabel("Residuals")
```

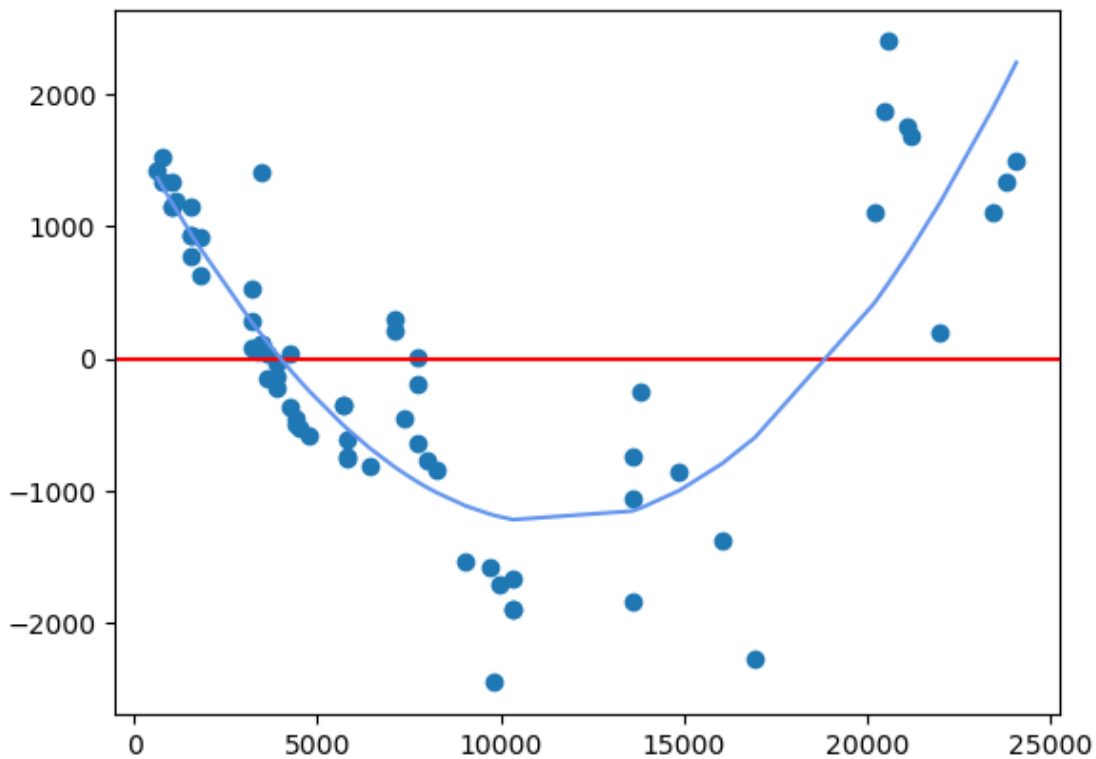
[76]: Text(0, 0.5, 'Residuals')



The graph below plots the residuals against Clarity. All residuals are negative for Clarity=2 and Clarity=8. There are more positive than negative residuals for Clarity=3, Clarity=4, and Clarity=5. All residuals are negative for Clarity 6 except for one large positive outlier. This is strong evidence that the model has not adequately captured the relationship in the data.

[77]:
```
plt.scatter(Diamonds_211["Clarity"], resid)
plt.axhline(0, color="red")
plt.title("Residuals Versus Clarity")
plt.xlabel("Clarity")
plt.ylabel("Residuals")
```

[77]: Text(0, 0.5, 'Residuals')



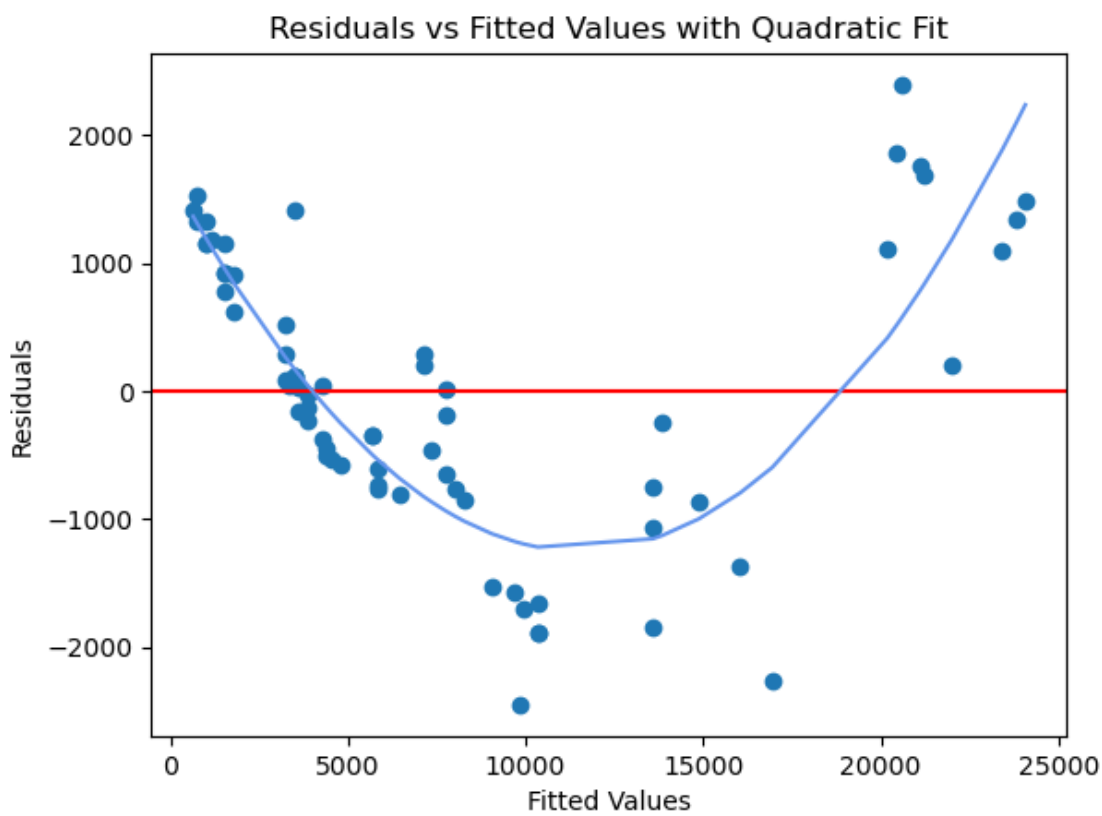
[ ]:

```
```

To test functional form, we will use the Ramsey test. The graph below illustrates the logic underlying the Ramsey test. For this illustration, I chose the diamonds with Clarity 5. There are 69 diamonds with Clarity=5. The first command selects the diamonds that have Clarity 5 and places them in a dataframe named Diamonds_69. The second command estimates a regression of Price versus Carat for those diamonds and names the result RegDiam69. The third and fourth commands calculate the fitted values and residuals for those 69 diamonds and names the results yhat and resid. The fifth command calculates the squared yhat values and names the result yhatsq. The sixth command plots the residuals versus the fitted values and the seventh puts a red horizonal command at zero on the graph. The eighth command places resid, yhat, and yhatsq in a dataframe named df. The ninth command estimates a regression of resid vs yhat and yhatsq and names the result ResidReg. The tenth command extracts the fitted values from ResidReg and names the result res_hat. The eleventh command adds res_hat to the graph creating the curve in the graph. The remaining three commands label the plot. If the estimated model fits the data well, the solid curve differ little from a horizontal line. In this instance, the estimated model fits the data poorly. If the model fits the data well, the coefficients of yhat and yhatsq in ResidReg should not differ significantly. The p-value for a test of the null hypothesis that both coefficients are zero is reported by the Ramsey test. The Ramsey test also estimates a regression with yhat, the square of yhat, and the cube of yhat. It then reports the p-value for the null hypothesis that all three of the coefficients are zero. Both p-values are reported by the Ramsey test.

[78]:
```
Diamonds_69 = Diamonds_211.query("Clarity == 5")
RegDiam69 = tprstats.model("cs", "Price~Carat", Diamonds_69)
resid = RegDiam69.resid
yhat = RegDiam69.predict()
yhatsq = yhat**2
plt.scatter(yhat, resid)
plt.axhline(0, color="red")
df = pandas.DataFrame({"yhat": yhat, "yhatsq": yhatsq, "resid": resid})
ResidReg = tprstats.model("cs", "resid~yhat+yhatsq",df)
res_hat = ResidReg.predict()
plt.plot(yhat, res_hat, color="cornflowerblue")
plt.title("Residuals vs Fitted Values with Quadratic Fit")
plt.xlabel("Fitted Values")
plt.ylabel("Residuals")
```

[78]: Text(0, 0.5, 'Residuals')



The command below conducts the Ramsey test for RegDiam69. Behind the scenes, the Ramsey test estimates the ResReg model above. It then reports the test of the null hypothesis that the coefficients that the coefficients of yhat and yhatsq are both zero. Then it estimates a regression of resid against yhat and the square and cube of yhat. It then tests the null hypothesis that all three

coefficients are zero and reports the p-value The low p-values confirm that the model RegClarity5 fits the diamond data poorly.

[79]:
```
RegDiam69.ramsey_test()
```

[79]:
```
   power        pvalue
0      2  5.153924e-19
1      3  4.118908e-18
```

Let's now conduct the Ramsey test for our regression with 211 diamonds. The low p-values below strongly reject our functional form. The Ramsey test is a diagnostic. It tests the functional form of the model we estimated, but it does not recommend an alternative model. We will see how functional form can be chosen in the next chapter.

[80]:
```
RegDiam211.ramsey_test()
```

[80]:
```
   power        pvalue
0      2  4.543472e-15
1      3  1.698258e-16
```

1.8 Three Useful Facts About Regression

The following properties hold whether or not the estimated model fits the data well.

1.8.1 Property 1: The estimated residuals always have a mean of zero

Why does this property hold? Suppose, hypothetically, that the mean of the residuals were positive. The sum of squared errors could then be reduced. How? Increase the constant term by an amount equal to the mean of the residuals.

[81]:
```
resid.mean()
```

[81]:
```
-0.0000
```

1.8.2 Property 2: Estimated residuals are uncorrelated with explanatory variables

This is confirmed below for our diamond regression.

Why does this property hold? Suppose, hypothetically, that the residuals were positively correlated with Carat. The sum of squared errors could then be reduced by increasing the coefficient of Carat. If the residuals were negatively correlated with Carat, the sum of squared errors could be reduced by reducing the coefficient of Carat.

[82]:
```
resid.corr(Diamonds_211["Carat"]), resid.corr(Diamonds_211["Clarity"])
```

```
/workspaces/tprstats-python-book/.pixi/envs/default/lib/python3.12/site-
packages/numpy/lib/_function_base_impl.py:3045: RuntimeWarning: invalid value
encountered in divide
  c /= stddev[:, None]
/workspaces/tprstats-python-book/.pixi/envs/default/lib/python3.12/site-
packages/numpy/lib/_function_base_impl.py:3046: RuntimeWarning: invalid value
```

```
encountered in divide
  c /= stddev[None, :]
```

[82]: (0.0000, nan)

1.8.3 Property 3: The estimated residuals are uncorrelated with the fitted values

This is confirmed below for our diamond regression.

Why does this property hold? yhat is a linear function of the explanatory variables. The residuals are uncorrelated with the explanatory variables. Hence, this property follows from Property 2 above.

[83]:
```
resid.corr(pandas.Series(yhat))
```

[83]: nan

1.9 Summary

In this this chapter, we stated and interpreted the Gauss Markov theorem. We then did an application to investigate how diamond prices vary with Carat and Clarity. We calculated and interpreted confidence intervals for coefficients, and undertook prediction. We introduced the Wald test which can be used to test a null hypothesis about one or more coefficients. The chapter also introduced a diagnostic for functional form named the Ramsey test which is named after its inventor, James Ramsey. The chapter concluded with a summary of some useful facts about regression.